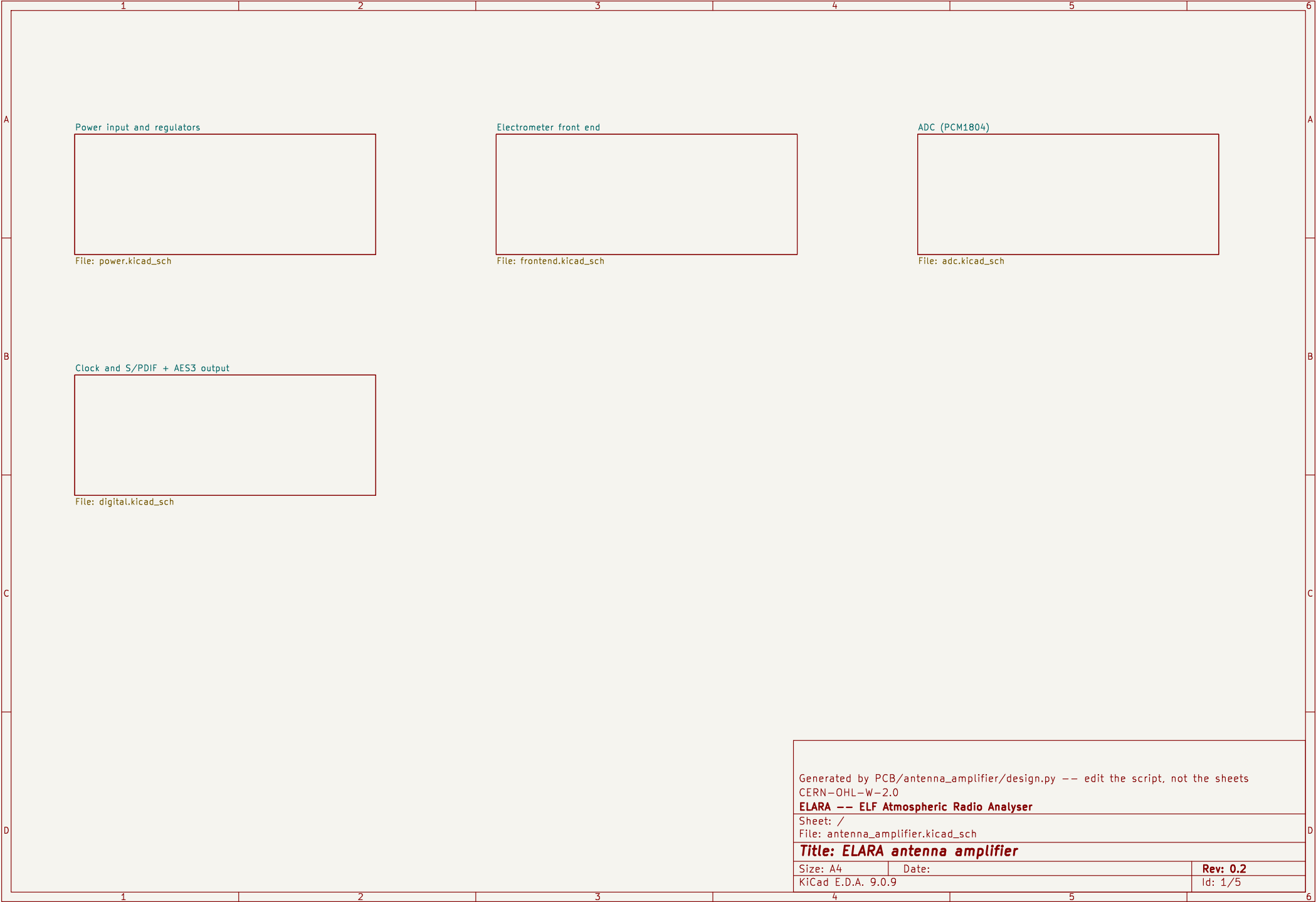
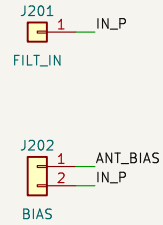


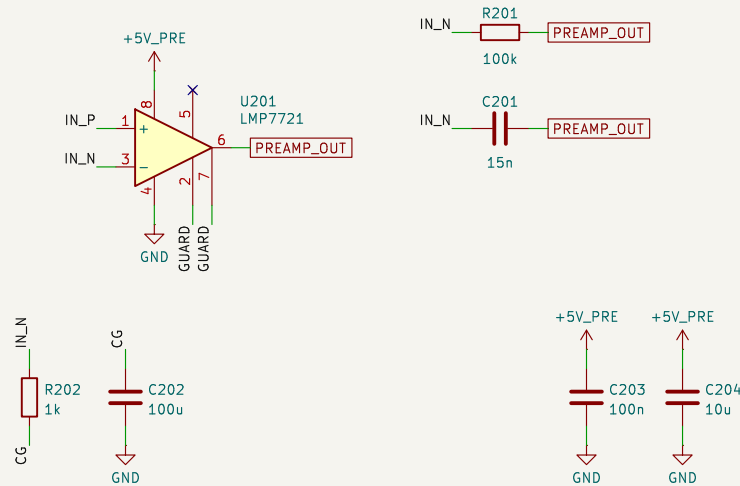


INPUT (comp. 1)

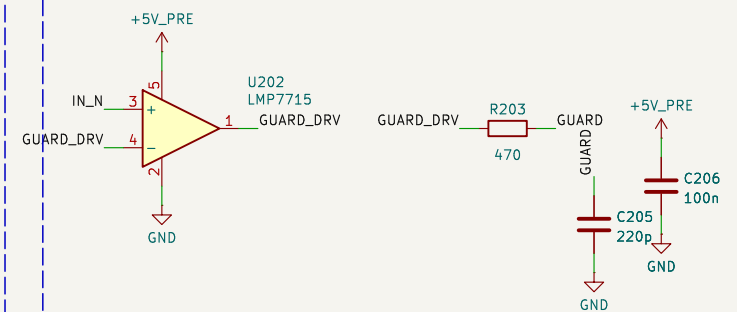


IN_P: guard on all layers,
no mask, PTFE turret.
J2: fit G—ohm bias R

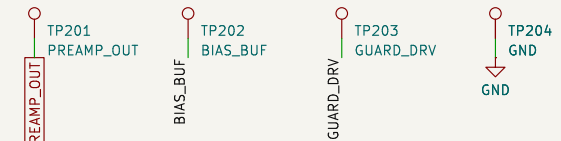
ELECTROMETER $G = 101$



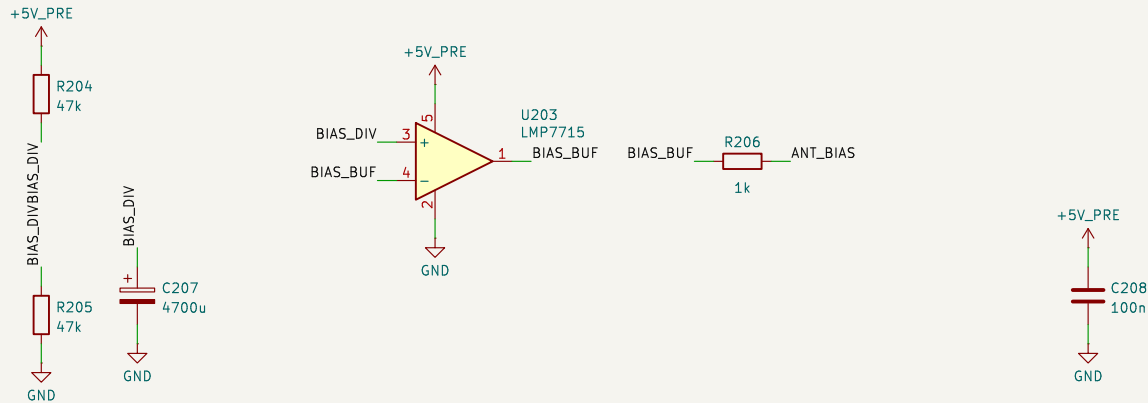
GUARD DRIVER



TEST POINTS



ANTENNA BIAS 2.5 V



Cg returns to GND: the 2.5 V buffer noise stays out of the gain path.
Guard buffer senses IN₋ (virtual short to IN₊): no bias current on IN_P.

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN—OHL—W—2.0

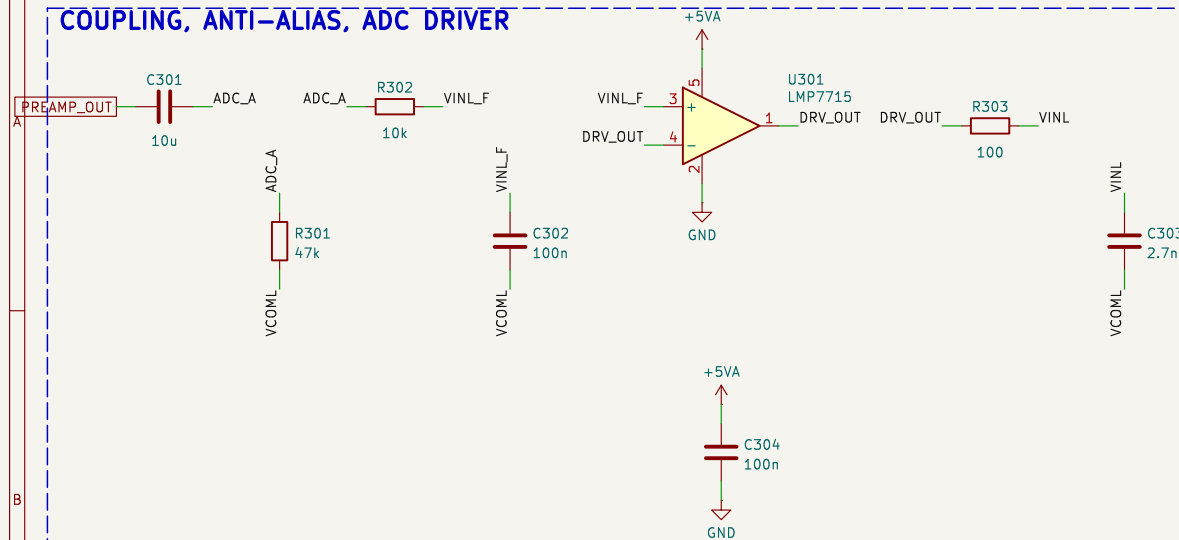
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Electrometer front end/
File: frontend.kicad_sch

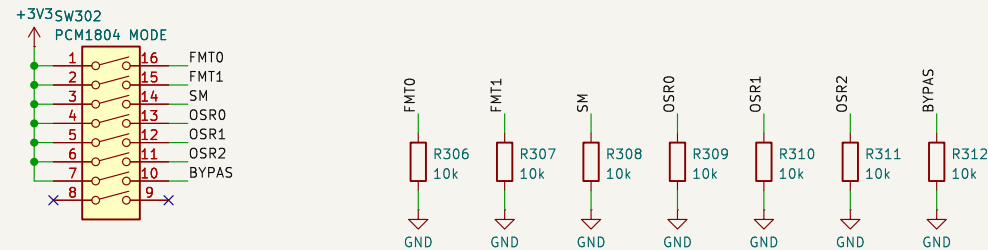
Title: ELARA antenna amplifier – Electrometer front end

Size: A4 Date: Rev: 0.2
KiCad E.D.A. 9.0.9 Id: 3/5

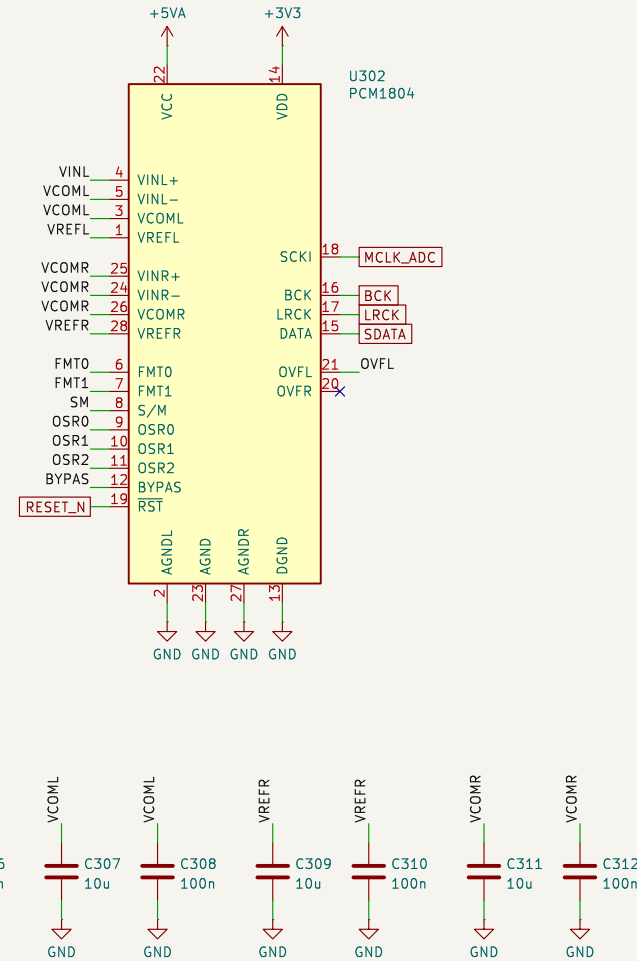
COUPLING, ANTI-ALIAS, ADC DRIVER



PCM1804 MODE (ON = 1)

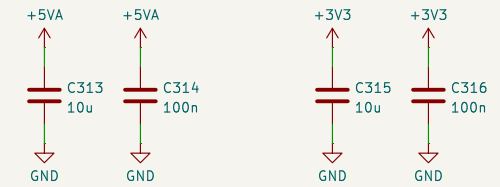


PCM1804 master, 192 kHz



Default (192 kHz): S/M=1 master, OSR2..0=111 (SCKI=128 fs), FMT1..0=01 I2S, BYPAS=1 (HPF off).
Change switches with power off, or press RESET afterwards.

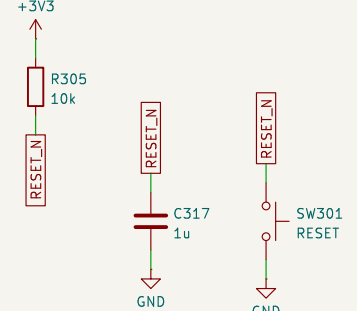
SUPPLY DECOUPLING



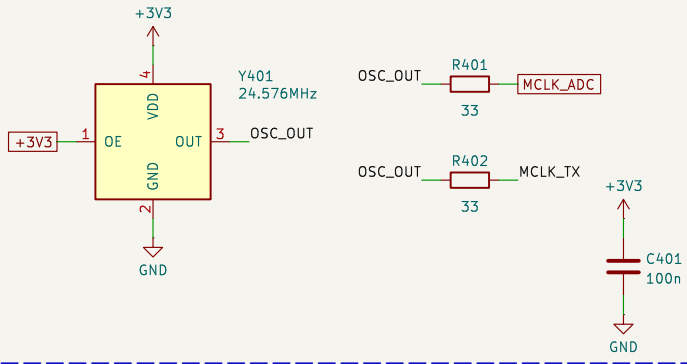
OVERLOAD



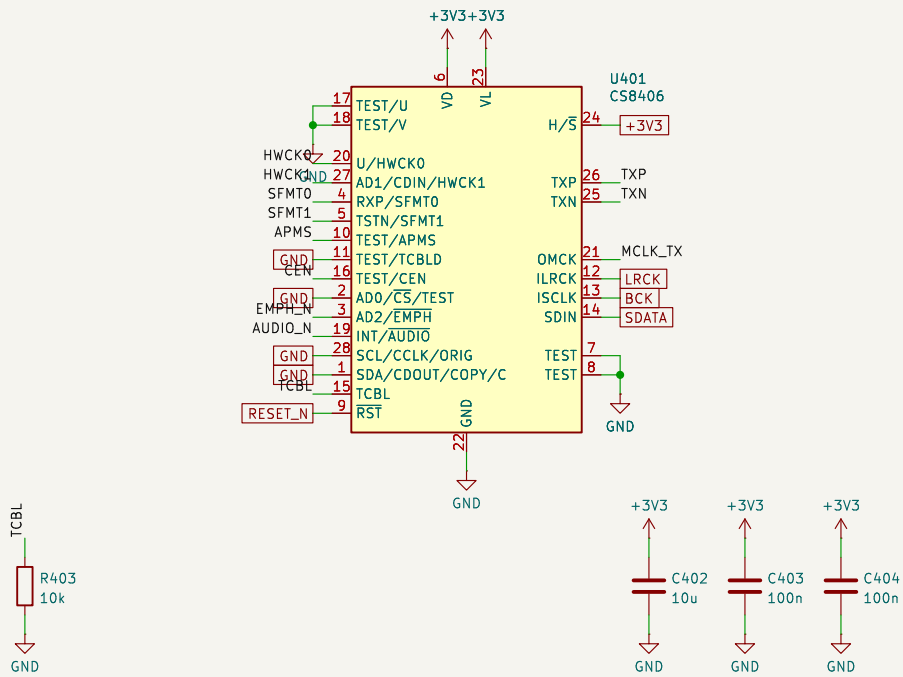
RESET



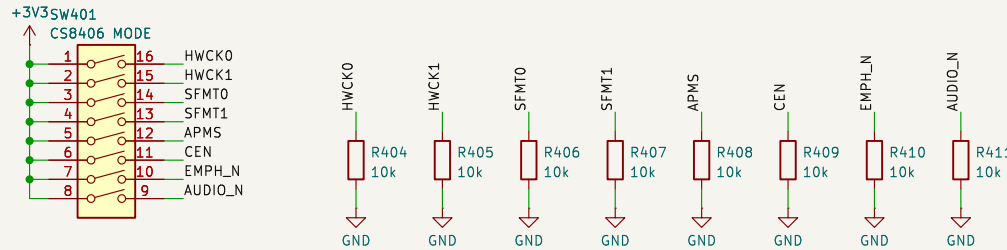
MASTER CLOCK 24.576 MHz



CS8406 hardware mode



CS8406 MODE (ON = 1)

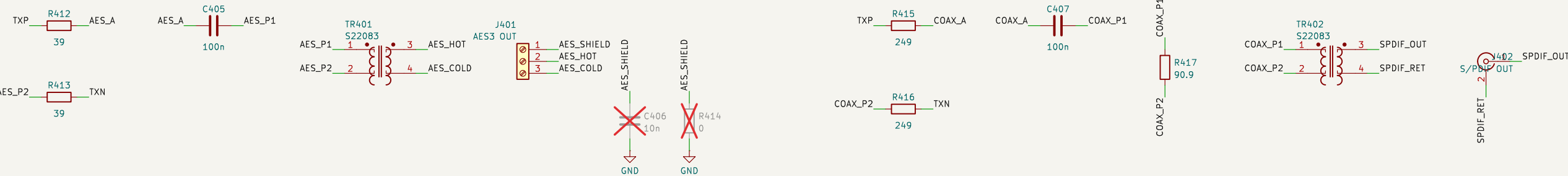


Default (192 kHz): HWCK -> OMCK = 128 fs, SFMT = 12S, APMS=0 (slave), CEN=0, EMPH_N=1, AUDIO_N=0.

OUTPUTS transformer isolated

AES3 110 R balanced

S/PDIF 75 R coax, 0.5 Vpp



AES3: 2 x 39 R + ~2 x 26 R driver ~ 110 R source, ~3 Vpp into 110 R.

S/PDIF: 2 x 249 R + 90.9 R shunt: 75 R source, 0.5 Vpp into 75 R.

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0

ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Clock and S/PDIF + AES3 output/
File: digital.kicad_sch

Title: ELARA antenna amplifier -- Clock and S/PDIF + AES3 output

Size: A3

Date:

Rev: 0.2

KiCad E.D.A. 9.0.9

Id: 5/5